OPTICS · SEMESTER 5 · MODULE 4

Prepare: Polarisation

Everything you need before you open the question paper, from Plus Two up to the syllabus (Book 2 §20; explained here from Ghatak ch. 22 and §24.5, plus Book-2 topics like the Nicol prism and Laurent's polarimeter).

File 1 of 3	**Prepare** (this file): learn the ideas
File 2	Model paper: PYQs + new questions
File 3	Answer key: full solutions

Each answer in File 3 starts with topic codes such as **T9** **P2** **M3** . If a code is new to you, read that topic here first.

How to read this file

Read one topic at a time, in order. Each topic builds only on the ones before it.

REMIND ME

Facts from Class 10, Plus One or Plus Two that you have probably forgotten.

MATHS TOOL

A small piece of maths, explained the first time it is needed.

THE IDEA

The physics in plain words, before any symbols.

RESULT

What you must remember and write in the exam.

EXAM TRAP

A common mistake, and how to avoid it.

CHECK YOURSELF

Questions with the answer printed just below. Cover it with your thumb first.

Colours and codes

blue	incoming light, or the electric field E
green	ordinary (O) ray, or light that passes
orange	extraordinary (E) ray, or reflected light
purple	optic axis, or the pass axis of a polariser
short lines	vibration in the plane of the page
dots	vibration at 90° to the page

Topic codes

P1–P4	Plus One / Plus Two bridge
M1–M4	Maths tools
T1–T12	Module 4 topics
Q12	question 12 of the model paper

The map of Module 4

SYLLABUS UNIT	TOPICS
17. Polarisation introduction	T1
18. Production of linearly polarised light	T3, T4, T7
19. Polariser and analyser	T2
20. Double refraction, Huygens' explanation	T5, T6
21. Wave plates	T8, T9
22. Production and analysis of polarised light	T10, T11, T12

15 of the 70 external marks. Past papers asked definitions, wave-plate thickness numericals (four times!), Brewster's law, Huygens' explanation, Laurent's polarimeter and 10-mark essays on producing and analysing circular and elliptical light.

The whole module in one story

THE IDEA

1. Light is a wave whose electric field shakes **sideways**, at 90° to its direction of travel. "Polarisation" is about *which* sideways direction (T1).

2. A polariser lets through only the part of the field along its axis. Two polarisers give Malus' law (T2).

3. We can make polarised light by absorption, scattering, reflection at Brewster's angle, or by double refraction in a crystal (T3, T4, T7).

4. In calcite, light splits into two rays with different speeds and perpendicular vibrations. Huygens explained it with two kinds of wavelets (T5, T6).

5. Two perpendicular vibrations with a phase difference make a line, a circle or an ellipse (T8). A crystal plate creates exactly that phase difference: wave plates (T9).

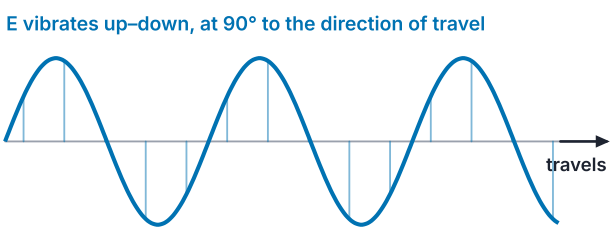
6. So we can produce and identify every kind of polarised light (T10, T11). Some materials even rotate the vibration: optical activity (T12).

P1 Waves: the words you need

> Amplitude, phase, intensity, and the difference between transverse and longitudinal.

REMIND ME · PLUS ONE CH 14 (WAVES)

- A wave: $y = a\cos(kz - \omega t)$. a = amplitude (biggest shake), λ = wavelength, $k = 2\pi/\lambda$, $\omega = 2\pi\nu$.
- The angle inside the cosine is the **phase**. Two waves whose phases differ by δ are "out of step by δ".
- A path difference Δ gives a phase difference $\delta = \frac{2\pi}{\lambda}\Delta$. So $\lambda/4$ of path = $\pi/2$ (90°) of phase, and $\lambda/2$ of path = π (180°).
- **Intensity ∝ (amplitude)²**. Double the amplitude, four times the brightness.



Light is a **transverse** wave: the electric field vibrates perpendicular to the direction the light travels. Only transverse waves can be polarised (sound in air, a longitudinal wave, cannot).

TRANSVERSE VS LONGITUDINAL

Transverse: the shaking is at 90° to the direction of travel (a rope wave, light). **Longitudinal**: the shaking is along the direction of travel (sound in air). A transverse wave can shake up–down, left–right or anything in between, so it has a "direction of shaking" to choose. That choice is polarisation. A longitudinal wave has no such choice, so it cannot be polarised.

CHECK YOURSELF

A path difference of 3λ/4. Phase difference?

$\frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4} = \frac{3\pi}{2}$ (270°).

P2 Light is an electromagnetic wave

Which arrow is "the light"? The electric field.

REMIND ME · PLUS TWO CH 8

- Light consists of an electric field $\vec{E}$ and a magnetic field $\vec{B}$, both oscillating.
- $\vec{E} \perp \vec{B}$, and both are perpendicular to the direction of travel. So light is **transverse**.
- Speed in vacuum $c = 1/\sqrt{\mu_0 \varepsilon_0} = 3 \times 10^8$ m/s; in a medium $v = c/n$.

THE IDEA

When we talk about the "vibration" of light we mean the **electric field** $\vec{E}$, because it is $\vec{E}$ that pushes the electrons in our eyes, films and detectors. The polarisation of light = the direction in which $\vec{E}$ vibrates.

CHECK YOURSELF

Light travels along z with $\vec{E}$ along x. Along which axis is $\vec{B}$?

Along y: perpendicular to both $\vec{E}$ and the direction of travel.

P3 Refraction, total internal reflection, Huygens' principle

Three Plus Two ideas that this module reuses.

REMIND ME · CLASS 10 / PLUS TWO CH 9, 10

- **Snell's law**: $n_1 \sin i = n_2 \sin r$. Angles from the normal. Refractive index $n = c/v$.
- **Total internal reflection**: going from a denser to a rarer medium ($n_1 > n_2$), if the angle of incidence is more than the **critical angle** C, where $\sin C = n_2/n_1$, no light goes out: all of it reflects. (Needed for the Nicol prism, T7.)
- **Huygens' principle**: every point on a wavefront acts as a source of small "secondary wavelets". After a time t, the new wavefront is the surface touching all the wavelets (their envelope). (Needed for T6.)

CHECK YOURSELF

Critical angle for light going from glass of $n = 1.658$ into a layer of $n = 1.55$?

$\sin C = 1.55/1.658 = 0.935$, so $C \approx 69.2^\circ$.

P4

What Plus Two already told you about polarisation

> NCERT XII §10.7 in one box. This module goes deeper into each line.

REMIND ME · PLUS TWO CH 10

- Ordinary light (sun, bulb) is **unpolarised**: its $\vec{E}$ points in random sideways directions that change very fast.
- A **polaroid** lets through only the vibrations along its "pass axis". The light that comes out is **plane (linearly) polarised**, with half the intensity.
- Two polaroids at angle θ: $I = I_0 \cos^2 \theta$ (Malus' law).
- Sunlight scattered by the sky at 90° is polarised.
- Light reflected at Brewster's angle ($\tan i_B = n$) is polarised.

Good news: all of this is still true. Now you will learn *why*, and meet new tools: crystals that split light (calcite), plates that turn a line of vibration into a circle, and liquids that rotate the vibration.

M1 Splitting an arrow into two parts

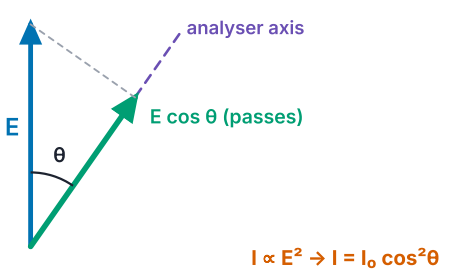
Components: the single most useful tool in this module.

REMIND ME · CLASS 10 TRIGONOMETRY

In a right triangle with hypotenuse E and angle θ: the side next to θ is $E\cos\theta$, the side opposite θ is $E\sin\theta$.

THE IDEA

Any arrow E can be replaced by two perpendicular arrows that add up to it: one along a chosen line ($E\cos\theta$) and one across it ($E\sin\theta$), where θ is the angle between the arrow and the line. Nothing is lost: $(E\cos\theta)^2 + (E\sin\theta)^2 = E^2$.



Why $\cos^2$: split E into a part along the analyser's axis ($E\cos\theta$, which passes) and a part across it (blocked). Intensity goes as amplitude squared.

CHECK YOURSELF

E = 10 units at 60° to a line. Components along and across it?

Along: $10\cos 60° = 5$. Across: $10\sin 60° = 8.66$.
Check: $25 + 75 = 100 = 10^2$. ✓

M2 Trig identities you will use

> Only five. Each with a number check.

RESULT

$$\sin^2\theta + \cos^2\theta = 1$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$\cos^2\theta = \tfrac{1}{2}(1 + \cos 2\theta)$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

1. Check $\sin 2\theta$ at θ = 30°: $\sin 60° = 0.866$; $2 \times 0.5 \times 0.866 = 0.866$ ✓

2. Check $\cos^2$ at θ = 60°: $\cos^2 60° = 0.25$; $\frac{1}{2}(1 + \cos 120°) = \frac{1}{2}(1 - 0.5) = 0.25$ ✓

REMIND ME · RADIANS

π rad = 180°. So $\pi/2$ = 90°, $\pi/4$ = 45°, $\pi/3$ = 60°.

M3 Two vibrations, a phase difference, and an ellipse

> The maths behind circular and elliptical light (T8).

REMIND ME · PLUS ONE SHM AND CONIC SECTIONS

Simple harmonic motion: $x = a\cos\omega t$. The point moves back and forth between $-a$ and $+a$.

Circle of radius a: $x^2 + y^2 = a^2$. Ellipse with half-axes a (along x) and b (along y): $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

THE IDEA

A point is pushed along x by $x = a\cos\omega t$ and along y by $y = b\cos(\omega t - \delta)$ at the same time. What path does it trace? Get rid of t and you get the path. For $\delta = \pi/2$: $y = b\cos(\omega t - \frac{\pi}{2}) = b\sin\omega t$ (because $\cos(A - 90^\circ) = \sin A$). Then $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2\omega t + \sin^2\omega t = 1$: an ellipse (a circle if $a = b$).



δ = 0 δ = π/4 δ = π/2 δ = 3π/4 δ = π

x = a cos ωt, y = a cos(ωt − δ): line → ellipse → circle → ellipse → line

Adding two perpendicular vibrations of equal amplitude with phase difference δ. δ = 0 or π: a straight line (linear). δ = π/2: a circle (circular). In between: ellipses. With unequal amplitudes, δ = π/2 gives an ellipse with axes along x and y.

CHECK YOURSELF

$x = 3\cos\omega t$, $y = 3\cos\omega t$ ($\delta = 0$). Path?

$y = x$ always: a straight line at 45°.

M4 Small numbers: Å, nm, µm

> Wave-plate numericals are pure unit-handling. Get this right and they are free marks.

1 Å (ångström)	10^{-10} m
1 nm	10^{-9} m = 10 Å
1 µm	10^{-6} m = 0.001 mm
5500 Å	$5500 \times 10^{-10} = 5.5 \times 10^{-7}$ m
589 nm	5.89×10^{-7} m

1. Typical sum: $d = \dfrac{\lambda}{4(\mu_e - \mu_o)} = \dfrac{5.5 \times 10^{-7}}{4 \times 0.01}$
   Write λ in metres first.

2. $= \dfrac{5.5 \times 10^{-7}}{0.04} = 1.375 \times 10^{-5}$ m
   Dividing by 0.04 is multiplying by 25.

3. $= 13.75$ µm $= 0.01375$ mm
   10^{-5} m = 10 µm.

EXAM TRAP

Forgetting to convert Å to m gives an answer 10^{10} times too big. Always write the unit at every step.

UNITS 17–19

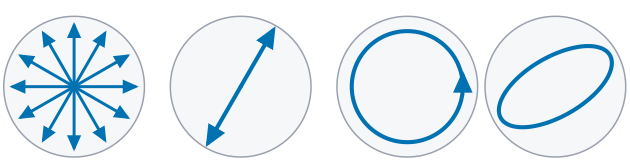
What polarisation is

T1 Polarisation and its kinds

> The definition (PYQ N24 Q10) and the three kinds you must tell apart (PYQ N20 Q34).

RESULT · DEFINITION

Polarisation is the property of a transverse wave by which its vibrations (for light, the electric field $\vec{E}$) are confined to a particular direction, or follow a definite pattern, in the plane perpendicular to the direction of travel. Light whose $\vec{E}$ keeps a fixed pattern is **polarised**. Light whose $\vec{E}$ changes direction randomly and rapidly is **unpolarised**.



States of polarisation, drawn as the path traced by the tip of the electric field vector. Unpolarised light changes direction randomly and very fast; polarised light keeps an orderly pattern.

Plane (linear)	$\vec{E}$ vibrates along one fixed line.
Circular	The tip of $\vec{E}$ goes round a circle (constant length, steadily turning).
Elliptical	The tip of $\vec{E}$ traces an ellipse.
Unpolarised	$\vec{E}$ jumps between all sideways directions, randomly, about 10^8 times a second. Drawn as a star of arrows, or as lines + dots.
Partially polarised	A mixture of unpolarised and polarised light.

PLANE OF VIBRATION VS PLANE OF POLARISATION

The **plane of vibration** contains $\vec{E}$ and the direction of travel. The **plane of polarisation** is the plane perpendicular to it (it contains $\vec{B}$). This old naming confuses everyone: when in doubt, just say "$\vec{E}$ vibrates along ...".

EXAM TRAP

Circular light is not "rays going round in a spiral". The light goes straight; at any one point the **arrow** $\vec{E}$ turns round, keeping the same length.

CHECK YOURSELF

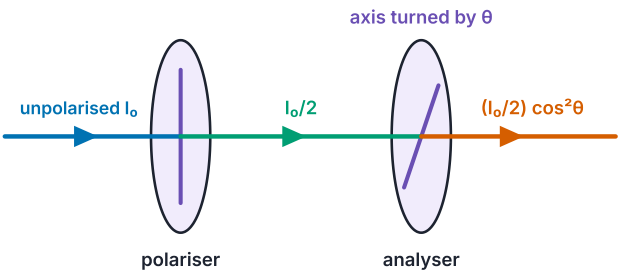
Can sound waves in air be polarised? Why?

No. Sound in air is longitudinal: the air shakes along the direction of travel, so there is no sideways direction to choose.

In the model paper: **Q1** (PYQ N24 Q10), **Q2**, **Q3**, **Q4**, **Q40**.

T2 Polariser, analyser and Malus' law

> What happens to the brightness as you turn one polariser behind another.



Polariser and analyser. The polariser passes half of unpolarised light. If the analyser's axis makes angle θ with the polariser's, only the component $E\cos\theta$ gets through, so the intensity becomes $I\cos^2\theta$ (**Malus' law**). Crossed (θ = 90°): darkness.

THE IDEA

A **polariser** has a **pass axis**. It lets through only the part (component, M1) of $\vec{E}$ along that axis and blocks the part across it.

Unpolarised light points every way equally, so on average exactly **half** the intensity gets through, whatever the angle.

A second polariser used to test the light is called the **analyser**.

Malus' law, step by step

1. After the polariser: plane polarised light, amplitude E_0, intensity $I_0 \propto E_0^2$

2. Analyser axis at angle θ to that vibration. Part passed: $E_0 \cos\theta$

   Component along the pass axis (M1). The part $E_0 \sin\theta$ is blocked.

3. $I \propto (E_0 \cos\theta)^2$, so $I = I_0 \cos^2\theta$

   Intensity ∝ amplitude² (P1).

RESULT · MALUS' LAW

$$I = I_0 \cos^2 \theta$$

θ = 0°: all passes. θ = 90° ("crossed"): darkness. Turning the analyser through 360° gives two bright and two dark positions.

Worked example: three polaroids

1. Unpolarised I_0 through polaroid 1: $\frac{I_0}{2}$
   Unpolarised always halves.
2. Polaroid 2 at 45° to polaroid 1: $\frac{I_0}{2} \cos^2 45^\circ = \frac{I_0}{4}$
3. Polaroid 3 at 90° to polaroid 1, so 45° to polaroid 2: $\frac{I_0}{4} \cos^2 45^\circ = \frac{I_0}{8}$
   Always use the angle between **neighbouring** polaroids.

Surprise: without the middle polaroid, crossed polaroids give darkness. Adding one in between lets light through! The middle one gives the light a new direction that has a component along the last axis.

PARTIALLY POLARISED LIGHT

Turn an analyser in it: the brightness changes between I_{max} and I_{min} but never reaches zero. Degree of polarisation $P = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$ (1 for plane polarised, 0 for unpolarised).

EXAM TRAP

Malus' law applies to light that is **already polarised**. For unpolarised light, halve it first, then apply $\cos^2$.

In the model paper: **Q5** , **Q6** , **Q25** , **Q38** .

T3 Polaroids, wire grids and scattering

Three ways to remove one direction of vibration.

DICHROISM AND POLAROID

Some materials absorb one direction of vibration much more strongly than the other (tourmaline is a natural example). A **Polaroid** sheet has long chain molecules (with iodine) all lined up. Electrons can move easily along the chains, so $\vec{E}$ along the chains is absorbed. $\vec{E}$ across the chains passes. The pass axis is **perpendicular to the chains**.

WIRE GRID

Fine parallel wires, closer together than the wavelength, do the same for long waves (microwaves, infrared): $\vec{E}$ along the wires drives currents and is absorbed or reflected; $\vec{E}$ perpendicular to the wires passes.

WHERE THE "PICKET FENCE" PICTURE LIES

A rope through a fence of vertical slits passes vertical shakes. But a wire grid passes the vibration **perpendicular** to its wires: the opposite. And a fence cannot explain $\cos^2\theta$. Think "keeps the component along its pass axis", not "slits".

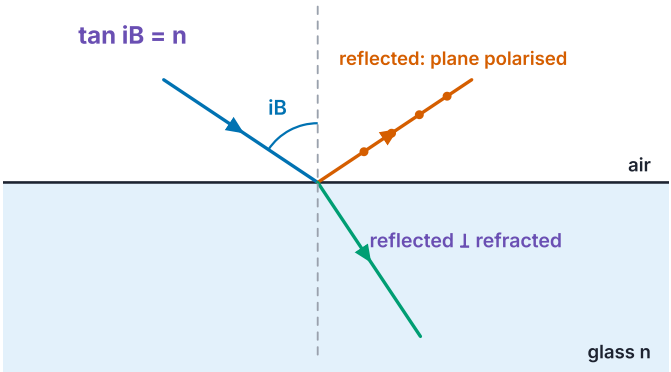
SCATTERING

Sunlight makes the electrons of air molecules shake like tiny antennas. A shaking charge sends out no light along its own line of shaking. So light scattered at 90° to the sunbeam can only contain the vibration perpendicular to both the beam and the viewing line: it is plane polarised. This is why the blue sky, seen at 90° from the sun, is polarised.

In the model paper: **Q7** , **Q8** .

T4 Polarisation by reflection: Brewster's law

> PYQ N20 Q17: "State Brewster's law and what is Brewster window?"



Brewster's law. At the angle i_B with $\tan i_B = n$, the reflected light is completely plane polarised (vibrations perpendicular to the plane of incidence: the dots), and the reflected and refracted rays are at 90°.

RESULT · BREWSTER'S LAW

When unpolarised light falls on a transparent surface at the angle i_B given by

$$\tan i_B = n \quad \left(= \frac{n_2}{n_1}\right),$$

the reflected light is completely plane polarised, vibrating perpendicular to the plane of incidence. At this angle the reflected and refracted rays are at 90° to each other.

Why the two rays are at 90°, and why $\tan i_B = n$

1. Experiment (and wave theory) shows that at the polarising angle the reflected and refracted rays are perpendicular: $i_B + r = 90^\circ$

   Physical reason: in the glass, electrons shake along the refracted ray's $\vec{E}$ in the plane of incidence. That direction is along the would-be reflected ray, and a shaking charge sends nothing along its own line, so that vibration is missing from the reflected light.

2. So $r = 90^\circ - i_B$ and $\sin r = \cos i_B$

3. Snell: $n_1 \sin i_B = n_2 \sin r = n_2 \cos i_B$

4. $\tan i_B = \dfrac{n_2}{n_1}$

   Divide by $\cos i_B$. For air → glass of $n = 1.5$: $i_B = \tan^{-1} 1.5 = 56.3^\circ$. Air → water: 53.1°.

PILE OF PLATES AND THE BREWSTER WINDOW

At i_B only about 15% of the perpendicular vibration is reflected per surface; the transmitted beam is only **partly** polarised. A **pile of plates** (10–15 glass plates at i_B) reflects away a little more perpendicular light at every surface, so the transmitted beam becomes nearly plane polarised (vibrating in the plane of incidence).

A **Brewster window** is a glass window tilted at Brewster's angle, used at the ends of gas laser tubes. Light vibrating in the plane of incidence passes it with no reflection loss at all, so the laser gives polarised light.

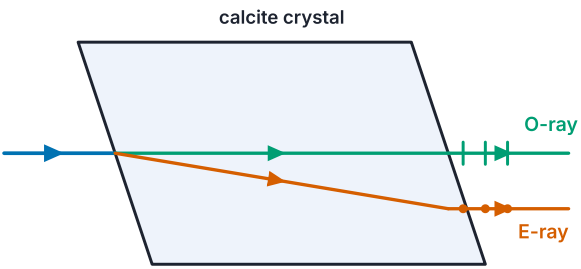
EXAM TRAP

The **reflected** beam is fully polarised; the **transmitted** beam is only partly polarised (unless you use a pile of plates).

In the model paper: **Q9** (PYQ N20 Q17), **Q10**, **Q26**, **Q27**.

T5 Double refraction: O-ray and E-ray

> Look at a dot through calcite and you see two dots. Why?



two rays, polarised at 90° to each other

Double refraction. Inside calcite one ray splits into two: the **ordinary** ray (obeys Snell's law, same speed in all directions) and the **extraordinary** ray (speed depends on direction). They come out plane polarised at right angles to each other.

RESULT · DOUBLE REFRACTION (PYQ N24 Q18)

When light enters certain crystals (calcite, quartz), it splits into two refracted rays:

- the **ordinary (O) ray**, which obeys Snell's law and has the same speed c/μ_o in every direction;
- the **extraordinary (E) ray**, which in general does not obey Snell's law and whose speed depends on its direction.

Both rays are plane polarised, with vibrations perpendicular to each other. This is **double refraction** (birefringence).

OPTIC AXIS AND PRINCIPAL SECTION

Optic axis: a *direction* in the crystal along which the O and E rays travel with the same speed, so there is no double refraction. (Any line parallel to it is also an optic axis.)

Uniaxial crystals (calcite, quartz) have one optic axis; **biaxial** crystals (mica) have two.

Principal section: a plane containing the optic axis and perpendicular to a pair of opposite faces. The E-ray vibrates in the principal section; the O-ray vibrates perpendicular to it.

NEGATIVE AND POSITIVE CRYSTALS

Negative (calcite)	$\mu_e < \mu_o$, so the E-ray is **faster** (off the axis). Calcite: $\mu_o = 1.658$, $\mu_e = 1.486$.
Positive (quartz)	$\mu_e > \mu_o$, so the E-ray is **slower**. Quartz: $\mu_o = 1.544$, $\mu_e = 1.553$.

RESULT · E-RAY SPEED IN A DIRECTION Θ FROM THE OPTIC AXIS

$$\frac{1}{v_e^2(\theta)} = \frac{\cos^2\theta}{v_\parallel^2} + \frac{\sin^2\theta}{v_\perp^2}$$

$v_\parallel = c/\mu_o$ along the axis (same as O), $v_\perp = c/\mu_e$ at 90° to it. (This is the polar equation of the E-wavelet ellipse, T6.)

1. PYQ N25 Q16: $v_{\parallel} = 1.81 \times 10^8$, $v_{\perp} = 2.02 \times 10^8$ m/s, θ = 45°: $\frac{1}{v_e^2} = \frac{1}{2}\left(\frac{1}{1.81^2} + \frac{1}{2.02^2}\right) \times 10^{-16}$

2. $= \frac{1}{2}(0.3052 + 0.2451) \times 10^{-16} = 0.2752 \times 10^{-16}$, so $v_e = 1.906 \times 10^8$ m/s
   Between the two extremes, as it must be.

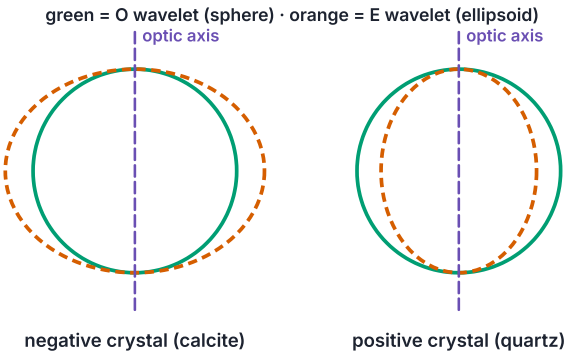
EXAM TRAP

"In a negative crystal the O-ray is slower than the E-ray": **true** (except along the optic axis, where they are equal). PYQ N22 Q9.

In the model paper: **Q11** (PYQ N19 Q8), **Q12**, **Q13** (PYQ N22 Q9), **Q14**, **Q29** (PYQ N25 Q16).

T6 Huygens' explanation of double refraction

> PYQ N24 Q9. Two kinds of wavelets instead of one.

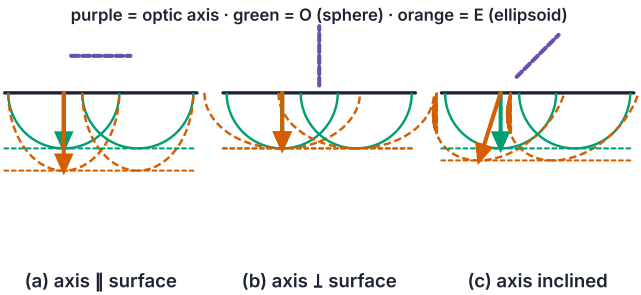


Huygens' explanation. Each point of the wavefront sends out two wavelets: a sphere (O) and an ellipsoid of revolution (E). They touch along the optic axis, where both rays have the same speed. Negative crystal: E is faster off-axis ($\mu_e < \mu_o$); positive crystal: E is slower.

RESULT · HUYGENS' THEORY

1. Every point of a wavefront entering a uniaxial crystal sends out **two** secondary wavelets.
2. The **O-wavelet is a sphere** (same speed in all directions).
3. The **E-wavelet is an ellipsoid of revolution** about the optic axis (speed depends on direction).
4. The sphere and the ellipsoid **touch along the optic axis**, where the speeds are equal.
5. Negative crystal: ellipsoid outside the sphere. Positive crystal: inside.
6. The new O and E wavefronts are the surfaces touching all the spheres and all the ellipsoids respectively. A ray goes from the wavelet's centre to its point of contact with the wavefront.

The important cases (light falling normally on the crystal)



Huygens' construction for normal incidence on calcite. (a) Both wavefronts parallel, rays along the same line, different speeds. (b) Wavefronts coincide: no double refraction. (c) The E-ray goes to the point where its wavefront touches the ellipsoid, so it is bent sideways.

Optic axis ⊥ surface	Sphere and ellipsoid touch along the direction of travel: both wavefronts coincide. **No double refraction**; one ray at speed c/μ_o.
Optic axis ∥ surface	Both wavefronts are parallel to the surface and both rays go straight on, **along the same line**, but at **different speeds**. No separation, but one falls behind the other: the basis of wave plates (T9).
Optic axis at an angle	The O-ray goes straight; the E-ray is bent sideways (it goes to the point where the

ellipsoid touches the wavefront), so you see two images.

OBLIQUE INCIDENCE, AXIS ⊥ PLANE OF INCIDENCE

Then the E-wavelet's cross-section in that plane is a circle too, of radius set by c/μ_e, so the E-ray obeys Snell's law with index μ_e: $\sin i / \sin r_e = \mu_e$.

EXAM TRAP

For the E-ray, the ray direction (centre → touching point) is in general **not** perpendicular to its wavefront.

CHECK YOURSELF

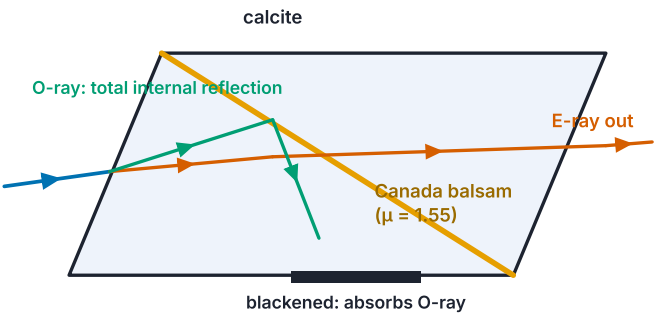
Light falls normally on calcite cut with its optic axis perpendicular to the faces. How many images?

One. Along the optic axis both waves travel at the same speed.

In the model paper: **Q15** (PYQ N24 Q9), **Q16** , **Q30** , **Q41** .

T7 The Nicol prism

> Using double refraction to make plane polarised light (Book 2 topic).



Nicol prism. A calcite crystal is cut along a diagonal and cemented with Canada balsam. The O-ray ($\mu_o = 1.658$) meets the balsam (1.55) beyond the critical angle and is totally reflected to the blackened side. The E-ray ($\mu_e = 1.486 < 1.55$) cannot be totally reflected and passes out, plane polarised.

1. **Construction:** a calcite crystal about three times as long as it is wide. Its end faces are ground from 71° to 68°. It is cut in two along a diagonal plane and the halves are cemented back with Canada balsam ($\mu = 1.55$)

2. **Splitting:** unpolarised light entering the first face splits into the O-ray ($\mu_o = 1.658$) and the E-ray ($\mu_e = 1.486$)

3. **O-ray removed:** it goes from calcite (1.658) towards balsam (1.55), denser → rarer. Critical angle $\sin C = 1.55/1.658$, so $C \approx 69^\circ$. It arrives at a larger angle, so it is totally internally reflected and absorbed by the blackened side
   Total internal reflection (P3).

4. **E-ray passes:** balsam (1.55) is denser than calcite for the E-ray (1.486), so it cannot be totally reflected. It comes out plane polarised, vibrating in the principal section

USES

As a **polariser** (makes plane polarised light) and as an **analyser** (rotate it in polarised light: two positions of darkness per turn). A Polaroid does the same job more cheaply but less perfectly.

In the model paper: **Q17** , **Q21** , **Q31** .

Circles and ellipses of light

T8 Two perpendicular vibrations

> Line, circle or ellipse: it all depends on the phase difference δ.

Let one vibration be along x and one along y, same frequency, phase difference δ:

$$E_x = a\cos\omega t, \qquad E_y = b\cos(\omega t - \delta)$$

1. $\cos\omega t = \dfrac{E_x}{a}, \quad \sin\omega t = \sqrt{1 - \dfrac{E_x^2}{a^2}}$
   From the first equation, and $\sin^2 + \cos^2 = 1$.

2. $\dfrac{E_y}{b} = \cos\omega t\cos\delta + \sin\omega t\sin\delta$
   Expand $\cos(\omega t - \delta)$ (M2).

3. $\dfrac{E_y}{b} - \dfrac{E_x}{a}\cos\delta = \sqrt{1 - \dfrac{E_x^2}{a^2}}\,\sin\delta$
   Substitute step 1, move the cos term across.

4. Square both sides: $\dfrac{E_y^2}{b^2} - \dfrac{2E_xE_y}{ab}\cos\delta + \dfrac{E_x^2}{a^2}\cos^2\delta = \sin^2\delta - \dfrac{E_x^2}{a^2}\sin^2\delta$
   $(p-q)^2 = p^2 - 2pq + q^2$ on the left; multiply out on the right.

5. Move the last term left and use $\cos^2\delta + \sin^2\delta = 1$: $\dfrac{E_x^2}{a^2} + \dfrac{E_y^2}{b^2} - \dfrac{2E_xE_y}{ab}\cos\delta = \sin^2\delta$
   The general ellipse. Time has disappeared, so this is the path of the tip of $\vec{E}$.



δ = 0 δ = π/4 δ = π/2 δ = 3π/4 δ = π

x = a cos ωt, y = a cos(ωt − δ): line → ellipse → circle → ellipse → line

Adding two perpendicular vibrations of equal amplitude with phase difference δ. δ = 0 or π: a straight line (linear). δ = π/2: a circle (circular). In between: ellipses. With unequal amplitudes, δ = π/2 gives an ellipse with axes along x and y.

RESULT · SPECIAL CASES

δ = 0	$E_y = \frac{b}{a}E_x$: a straight line → **plane** polarised
δ = π	$E_y = -\frac{b}{a}E_x$: a line the other way → **plane** polarised
δ = π/2, a = b	$E_x^2 + E_y^2 = a^2$ → **circularly** polarised
δ = π/2, a ≠ b	$\frac{E_x^2}{a^2} + \frac{E_y^2}{b^2} = 1$ → ellipse with axes along x and y
any other δ	a tilted ellipse → **elliptically** polarised

RIGHT AND LEFT

The tip goes round either clockwise or anticlockwise. In Ghatak's convention (looking at the oncoming light, wave along +z), $0 < \delta < \pi$ is **right**-handed and $\pi < \delta < 2\pi$ is **left**-handed. E.g. $E_x = a\cos\omega t$, $E_y = a\sin\omega t$ ($\delta = \pi/2$) is right circular. Always state your convention: books differ.

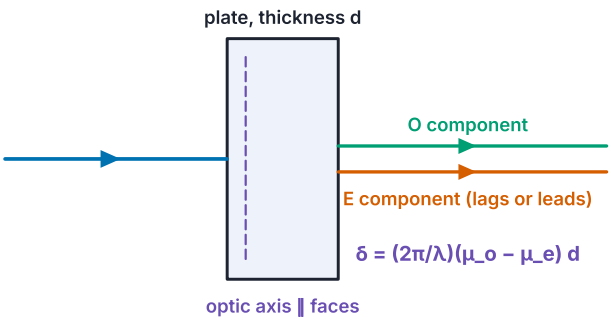
EXAM TRAP

δ = 0 gives a **line**, not a circle. A circle needs a quarter-period lag (π/2) **and** equal amplitudes.

In the model paper: **Q22** , **Q42** .

T9 Wave plates: quarter-wave and half-wave

> A slice of crystal that makes exactly the phase difference we want. Asked in five past papers.



A wave plate. Light enters at 90° to the optic axis, so the O and E vibrations travel along the same line but at different speeds. Leaving the plate, one is behind the other by a phase δ. Quarter-wave plate: δ = π/2. Half-wave plate: δ = π.

1. Cut a crystal plate with the optic axis **parallel** to its faces. Send plane polarised light in normally

   Huygens case "axis ∥ surface" (T6): O and E travel along the same line at different speeds.

2. Inside, the light is two perpendicular vibrations: E (along the axis) and O (across it), starting in step

3. Optical paths through thickness d: $\mu_o d$ and $\mu_e d$. Path difference $\Delta = (\mu_o - \mu_e)\, d$ (take the size)

4. Phase difference $\delta = \dfrac{2\pi}{\lambda}(\mu_o - \mu_e)\, d$

   P1: phase = $(2\pi/\lambda)\times$ path.

RESULT · PLATE THICKNESSES

Quarter-wave plate ($\Delta = \lambda/4$, $\delta = \pi/2$):

$$d = \frac{\lambda}{4(\mu_o - \mu_e)}$$

Half-wave plate ($\Delta = \lambda/2$, $\delta = \pi$):

$$d = \frac{\lambda}{2(\mu_o - \mu_e)}$$

Use the **size** of $(\mu_o - \mu_e)$; calcite gives a positive difference, quartz a negative one. Odd multiples of these thicknesses work too.

Worked example (PYQ N20 Q28; same idea as N19 Q30, N22 Q30, N25 Q17)

Quartz, λ = 5000 Å, $\mu_e = 1.553$, $\mu_o = 1.544$.

1. $|\mu_e - \mu_o| = 0.009$, λ = 5×10^{-7} m
   Convert Å to m (M4).

2. QWP: $d = \dfrac{5 \times 10^{-7}}{4 \times 0.009} = 1.39 \times 10^{-5}$ m $= 13.9$ μm

3. HWP: $d = \dfrac{5 \times 10^{-7}}{2 \times 0.009} = 2.78 \times 10^{-5}$ m $= 27.8$ μm
   Twice the QWP.

RESULT · WHAT THE PLATES DO

QWP, linear at 45° to axis	equal O and E parts, 90° apart → **circular**
QWP, linear at other angle	unequal parts, 90° apart → **elliptical** (axes along the plate's axes)
QWP, circular in	→ **linear** out (it undoes the quarter lag)
HWP, linear at angle φ	→ linear at −φ: the plane is **turned by 2φ**
HWP, circular in	→ circular of the **opposite** hand

EXAM TRAP

A wave plate is a QWP only for the one wavelength it was cut for. At another λ, δ is different and the light comes out elliptical.

In the model paper: **Q18** (PYQ N20 Q16), **Q19**, **Q20**, **Q28** (PYQ N20 Q28 + 3 more), **Q33**, **Q37**.

T10 Producing plane, circular and elliptical light

PYQ N20 Q23 and N19 Q34 (first half). A recipe for each.

RESULT · RECIPES

Plane polarised: pass ordinary light through a polariser (Nicol prism or Polaroid). Or reflect it at Brewster's angle.

Circularly polarised: plane polarised light from a Nicol falls on a **quarter-wave plate** with its optic axis at **45°** to the vibration. The O and E parts are equal ($E_0 \cos 45^\circ = E_0 \sin 45^\circ$) and leave a quarter-wave apart (δ = π/2): a circle.

Elliptically polarised: same, but with the optic axis at any angle φ other than 0°, 45° or 90°. Parts $E_0 \cos\phi$ and $E_0 \sin\phi$ are unequal, δ = π/2: an ellipse with axes along the plate's axes.

1. For the ellipse: $E_x = E_0 \sin\phi \cos\omega t$, $E_y = E_0 \cos\phi \cos(\omega t - \frac{\pi}{2})$
   Components (M1), quarter-wave lag.

2. $$\frac{E_x^2}{(E_0 \sin\phi)^2} + \frac{E_y^2}{(E_0 \cos\phi)^2} = 1$$
   T8 with δ = π/2. At φ = 45° the two denominators are equal: a circle.

PLATE BETWEEN CROSSED POLARISERS

Plate axis at φ to the first polariser, phase δ: $I = I_1 \sin^2 2\phi \, \sin^2(\delta/2)$, where I_1 is the intensity after the first polariser. QWP at 45°: $I_1/2$. HWP at 45°: all of I_1 passes.

In the model paper: **Q35** (PYQ N20 Q23), **Q32**, **Q39**.

T11 Analysing polarised light

> You are handed a beam. What kind is it? Use an analyser, then a QWP if needed.

1. **Rotate an analyser** in the beam.
   Three possible results: A, B or C below.

2. **A. Brightness drops to zero twice** per turn → **plane polarised**

3. **B. Brightness does not change** → unpolarised, circular, or a mix. Put a QWP before the analyser and rotate the analyser again: still no change → **unpolarised**; zero twice → **circular** (the QWP turned it into linear); varies but no zero → **unpolarised + circular** mix

4. **C. Brightness varies but never zero** → elliptical, or partly plane polarised. Set the QWP with its optic axis along the analyser's position of maximum, then rotate the analyser: zero twice → **elliptical**; no zero, maximum in the same place → **plane + unpolarised**; no zero, maximum moved → **elliptical + unpolarised**

WHY THE QWP TEST WORKS

A QWP adds a quarter-wave lag between its two axes. Circular light (or elliptical light lined up with the plate's axes) has a quarter-wave lag already. Adding another gives $\delta = 0$ or π: a line, which an analyser can extinguish. Unpolarised light has no fixed lag, so it stays unpolarised.

EXAM TRAP

Constant brightness through an analyser does **not** prove the light is unpolarised. Circular light does that too. You need the QWP test.

In the model paper: **Q39** (PYQ N19 Q34), **Q40**.

T12 Optical activity and Laurent's half-shade polarimeter

Liquids and crystals that turn the plane of vibration (PYQ N25 Q7, N20 Q5, N24 Q18).

RESULT · OPTICAL ACTIVITY

When plane polarised light passes through certain substances (quartz along its optic axis, sugar solution, turpentine), its plane of vibration is **rotated**. This is **optical activity**. Turning clockwise (looking towards the source) = **dextro-rotatory** (right-handed); anticlockwise = **laevo-rotatory**. The angle of rotation is proportional to the length of the path and (for solutions) to the concentration, and depends on wavelength and temperature.

FRESNEL'S EXPLANATION

Plane polarised light = right circular + left circular light of equal amplitude (their sideways parts cancel, the rest adds along a line). In an active substance the two circular waves travel at slightly different speeds ($n_L \neq n_R$). After a length z one is ahead of the other, and when they are added again the line has turned by

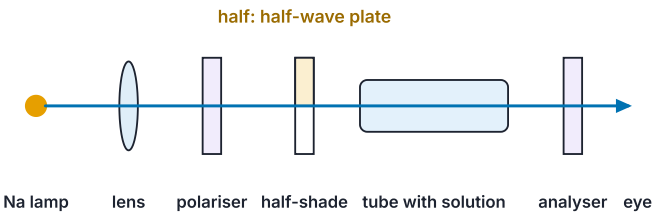
$$\phi = \frac{\pi}{\lambda}(n_L - n_R)\, z$$

RESULT · SPECIFIC ROTATION (PYQ N20 Q5)

The rotation produced by a 1 decimetre (10 cm) length of solution of concentration 1 g/cm³, at a given wavelength and temperature:

$$[\alpha]_{\lambda}^{T} = \frac{\theta}{l\,c} \qquad (\theta \text{ in degrees},\ l \text{ in dm},\ c \text{ in g/cm}^3)$$

Laurent's half-shade polarimeter (Book 2; PYQ N24 Q18)



Laurent's half-shade polarimeter. One half of the half-shade disc is a half-wave plate, the other half plain glass. With the analyser set for equal dimness of both halves, any rotation by the solution upsets the balance; turning the analyser back to equal dimness measures the rotation.

1. **Why a half-shade?** Finding the exact position of complete darkness by eye is not accurate: the brightness changes very slowly near zero. Comparing two halves of a field for **equal dimness** is far more sensitive

2. **The half-shade:** a disc, one half a half-wave plate (quartz, axis parallel to the dividing line), the other half plain glass that absorbs the same amount of light

3. The polariser's vibration makes a small angle ε with the plate's axis. In the glass half it stays at +ε; the HWP turns it to −ε (T9). So the two halves vibrate along OA and OB, symmetric about the axis

4. With the analyser exactly at 90° to the axis's bisector, both halves are **equally dim**. Turn it a little either way and one half brightens while the other darkens

5. **Measurement:** set equal dimness with water in the tube; note the analyser reading. Fill with the sugar solution: the balance is lost. Turn the analyser until equal dimness returns. The angle turned is the rotation θ

6. Specific rotation $[\alpha] = \dfrac{\theta}{l\,c}$, or with known $[\alpha]$, find the concentration $c = \dfrac{\theta}{[\alpha]\,l}$

CHECK YOURSELF

A 20 cm tube of sugar solution rotates the plane by 13.3°. $[\alpha] = 66.5^\circ$ per dm per g/cm³. Concentration?

$l = 2$ dm, $c = \dfrac{13.3}{66.5 \times 2} = 0.1$ g/cm³.

In the model paper: **Q23** (PYQ N25 Q7), **Q24** (PYQ N20 Q5), **Q34** , **Q36** (PYQ N24 Q18).

One-page summary

Polarisation	E confined to one direction or pattern ⊥ travel
Malus	$I = I_0 \cos^2\theta$; unpolarised → $I_0/2$ first
Brewster	$\tan i_B = n$; reflected ⊥ refracted; reflected fully polarised
Double refr.	O: obeys Snell, sphere. E: speed varies, ellipsoid. Both polarised, ⊥
Optic axis	direction of equal speeds, no splitting
Negative	calcite, $\mu_e < \mu_o$, E faster
Positive	quartz, $\mu_e > \mu_o$, E slower
E-ray speed	$\frac{1}{v_e^2} = \frac{\cos^2\theta}{v_\parallel^2} + \frac{\sin^2\theta}{v_\perp^2}$
Nicol	O-ray TIR at balsam (1.55); E passes
Ellipse	$\frac{E_x^2}{a^2} + \frac{E_y^2}{b^2} - \frac{2E_xE_y}{ab}\cos\delta = \sin^2\delta$
QWP	$d = \frac{\lambda}{4\lvert\mu_o - \mu_e\rvert}$, δ = π/2
HWP	$d = \frac{\lambda}{2\lvert\mu_o - \mu_e\rvert}$, δ = π; turns plane by 2φ
Circular	linear at 45° + QWP
Rotation	$\phi = \frac{\pi}{\lambda}(n_L - n_R)z$
Specific rot.	$[\alpha] = \theta/(lc)$, l in dm

Exam question → topics to know

PYQ	QUESTION (SHORT)	KNOW
N24 Q10	Define polarisation	T1 P1
N20 Q34	Plane/circular/elliptical; production; detect circular	T1 T4 T7 T10 T11
N20 Q17	Brewster's law, Brewster window	T4
N19 Q8	O and E rays plane polarised? (T/F)	T5
N22 Q9	Negative crystal: v_o < v_e? (T/F)	T5
N25 Q16	E-ray speed at 45°	T5 M2
N24 Q9	Huygens' explanation	T6 P3
N20 Q16, N22 Q23	Half-wave / quarter-wave plate	T9 T8
N19 Q30, N22 Q30, N20 Q28, N25 Q17	Plate thickness	T9 M4
N20 Q23, N19 Q34	Produce (and analyse) circular/elliptical	T10 T11 T8
N20 Q5, N25 Q7	Specific rotation; optical activity	T12
N24 Q18	Double refraction; Laurent polarimeter	T5 T12 T9